NAME: _____ CLASS: _____

BWAILA CLUSTER TERM I EXAMINATIONS

2023 MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION

CHEMISTRY

Subject Number: M038/II

Thursday, 1st December, 2022 **Time Allowed: 2 hour sessions**

10:00 am onwards

PAPER II

(40 marks)

PRACTICAL

Instructions

1. **This paper contains 5 printed pages. Please check.**
2. Before beginning, fill in your **name** at the top of each page of the question paper.
3. Write your answers on the question paper.
4. This paper consists of **two** sections **A** and **B**.
5. Section **A** consists of two descriptive questions on practical work to be answered in **1 hour**. Marks will be given for accurate and orderly presentation of facts supported by relevant diagrams.
6. In Section **B** there are **two** practical questions to be answered in **1 hour**.
7. Marks for Section **B** will be given for observation, accuracy and interpretation of results.
8. You should spend 30 minutes on each question. The 30 minute period allowed for each question includes 3 minutes to tidy up the apparatus and have it checked by the supervisor.
9. In the table provided on this page, **tick** against the question number you have answered.

Question Number	Tick if answered	Do not write in these columns		Examiner's Name
1				
2				
3				
4				

Turn over

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SECTION A (20 marks)

1. A student is being provided with a 0.1M sulphuric acid (H_2SO_4) and a 0.1M nitric acid (HNO_3). Describe an experiment on how to determine the strength of the acids using PH.

(10 marks)

2. Describe how the concentration of a solution can be determined using evaporation method.

(10 marks)

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SECTION B (20 marks)

3. You are provided with three test tubes in a rack, a measuring cylinder, distilled water, sodium hydroxide (NaOH) solution and solutions P, Q and R which are aluminum sulphate, zinc sulphate and copper sulphate but not necessarily in that order.

a. Pour 5cm3 of solution P into a clean test tube
b. Add 3 drops of sodium hydroxide solution into the test tube containing solution p.
c. Observe and record the colour of precipitate formed in an appropriate space in the table of results
d. Add more sodium hydroxide solution into the test tube until in excess and shake gently.
e. Observe and record the solubility of the precipitate as 'soluble' or insoluble in an appropriate space in the table of results
f. Rinse the measuring cylinder with distilled water
g. Repeat steps **a** to **f** using solutions Q and R

Table of Results

Solution	Colour of Precipitate formed after adding 4 drops of NaOH	Solubility of precipitate in excess NaOH
P		
Q		
R		

(6 marks)

h. Identify the cautions present in solution **P**, **Q** and **R.**

Solution **P**: _____

Solution **Q**: _____

Solution **R**: _____

(3 marks)

i. State any one source of error in the experiment.

_____

_____

(1 mark)

4. You are provided with a burette, a beaker, 2M sodium hydroxide (NaOH), Hydrochloric acid (HCl), measuring cylinder, a beaker, funnel, phenolphthalein indicator, dropper, clamp and clamp stand, white tile.

a. Put 10ml NaOH in a beaker

b. Arrange the set up as below

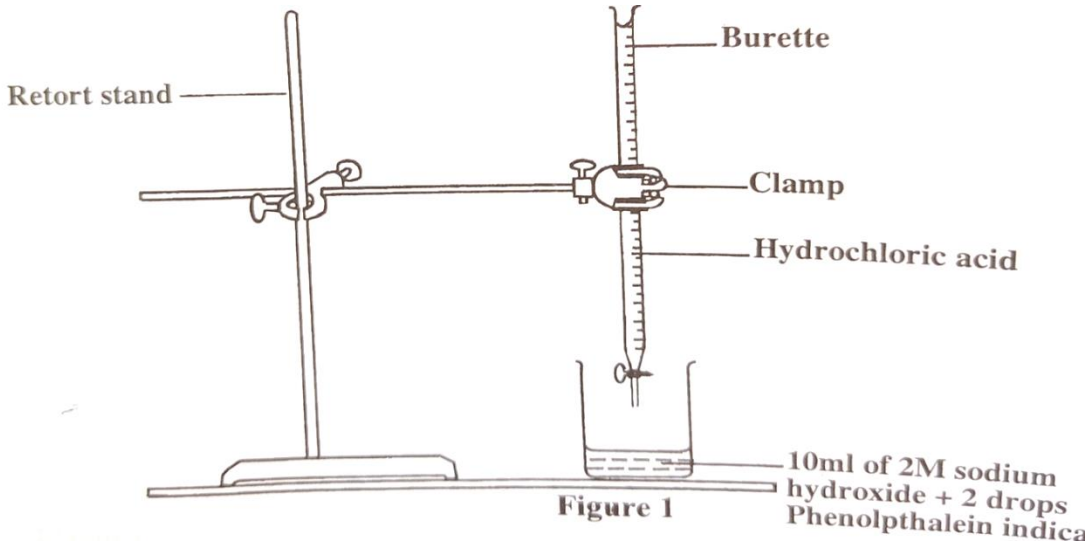

Figure 1

c. Add 5 drops of phenolphthalein indicator into the beaker.

d. Fill the burette to zero mark with hydrochloric acid (HCl).

e. Record initial volume of acid in **Table 2**.

Table 2

Initial volume of **HCl**	Final volume of **HCl**	**HCl used**

(3 marks)

f. Gradually add **HCl** into **0.1M NaOH** until endpoint is reached.

g. Record final volume of **HCl** in **Table 2**.

h. Calculate the concentration of acid used.

(5 marks)

i. Give any **two** sources of error from the experiment.

_____

_____

(2 marks)

END OF QUESTION PAPER